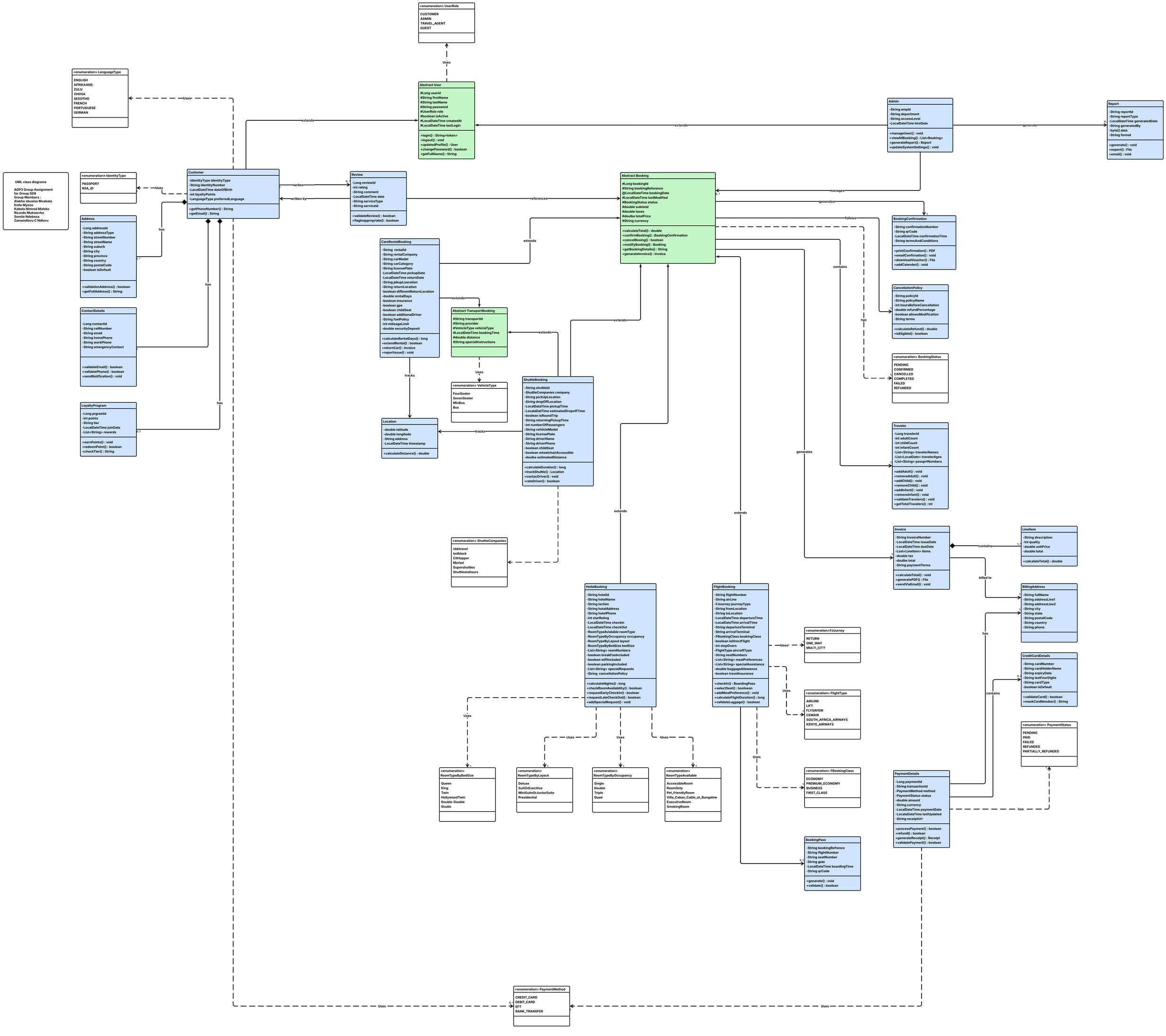
UML class diagrams
ADP3 Group Assignment
for Group SDB
Group Members :
Alakhe sbusiso Mxakato
Entle Myezo
Kabelo Nimrod Moloko
Ricardo Mukwevho
Somila Ndoboza
Zamandlovu C Ndlovu
<enumeration> UserRole
CUSTOMER
ADMIN
TRAVEL_AGENT
GUEST
Uses
<enumaration> LanguageType
ENGLISH
AFRIKAANS
ZULU
XHOSA
SESOTHO
FRENCH
PORTUGUESE
GERMAN
Uses
extends
Abstract User
#Long userId
#String firstName
#String lastName
#String password
#UserRole role
#boolean isActive
#LocalDateTime createdAt
#LocalDateTime lastLogin
+login() : String<token>
+logout() : void
+updateProfile() : User
+changePassword() : boolean
+getFullName() : String
extends
Admin
-String empId
-String department
-String accessLevel
-LocalDateTime hireDate
+manageUser() : void
+viewAllBooking() : List<Booking>
+generateReport() : Report
+updateSystemSettings() : void
generate
Report
-String reportId
-String reportType
-LocalDateTime generatedDate
-String generatedBy
-byte[] data
-String format
+generate() : void
+export() : File
+email() : void
<enumeration> IdentityType
PASSPORT
RSA_ID
Uses
Customer
-IdentityType identityType
-String identityNumber
-LocalDateTime dateOfBirth
-int layaltyPoints
-LanguageType preferredLanguage
+getPhoneNumber() : String
+getEmail() : String
writes
written by
Review
-Long reviewId
-int rating
-String comment
-LocalDateTime date
-String serviceType
-String serviceId
+validateReview() : boolean
+flagInappropriate() : boolean
references
Abstract Booking
#Long bookingId
#String bookingReference
@LocalDateTime bookingDate
#LocalDateTime lastModified
#BookingStatus status
#double subtotal
#double taxes
#doulbe totalPrice
#String currency
+calculateTotal() : double
+confirmBooking() : BookingConfirmation
+cancelBooing() : boolean
+modifyBooking() : Booking
+getBookingDetails() : String
+genarateInvoice() : Invoice
manages
generates
follows
contains
has
Address
-Long addressId
-String addressType
-String streetNumber
-String streetName
-String suburb
-String city
-String province
-String country
-String postalCode
-boolean isDefault
+validationAddress() : boolean
+getFullAddress() : String
has
ContactDetails
-Long contactId
-String cellNumber
-String email
-String homePhone
-String workPhone
-String emergencyContact
+validateEmail() : boolean
+validatePhone() : boolean
+sendNotification() : void
has
LoyaltyProgram
-Long prgramId
-int points
-String tier
-LocalDateTime joinDate
-List<String> rewards
+earnPoints() : void
+redeemPoint() : boolean
+checkTier() : String
has
BookingConfirmation
-String confirmationNumber
-String qrCode
-LocalDateTime confirmationTime
-String termsAndConditions
+printConfirmation() : PDF
+emailConfirmation() : void
+downloadVoucher() : File
+addCalender() : void
CancellationPolicy
-String policyId
-String policyName
-int hoursBeforeCancellation
-double refundPercentage
-boolean allowsModification
-String terms
+calculateRefund() : double
+isEligible() : boolean
<enumaration> BookingStatus
PENDING
CONFIRMED
CANCELLED
COMPLETED
FAILED
REFUNDED
CareRentalBooking
-String rentalId
-String rentalCompany
-String carModel
-String carCategory
-String licensePlate
-LocalDateTime pickupDate
-LocalDateTime returnDate
-String pikupLoacation
-String returnLocation
-boolean differentReturnLocation
-double rentalDays
-boolean insurance
-boolean gps
-boolean childSeat
-boolean additionalDriver
-String fuelPolicy
-int mileageLimit
-double securityDeposit
+calculateRentalDays() : long
+extendRental() : boolean
+returnCar() : invoice
+reportissue() : void
extends
extends
Abstract TransportBooking
#String transportId
#String provider
#VehicleType vehicleType
#LocalDateTime bookingTime
#double distance
#String specialInstructions
extends
Uses
<enumeration> VehicleType
FourSeater
SevenSeater
MinBus
Bus
tracks
Location
-double latitude
-double longitede
-String address
-LocalDateTime timestamp
+calculateDistance() : double
tracks
ShuttleBooking
-String shuttleId
-ShuttleCompanies company
-String picklpLocation
-String dropOffLocation
-LocalDataTime pickupTime
-LocalaDatTime estimatedDropoffTime
-boolean isRoundTrip
-String returningPickupTime
-int numberOfPassengers
-String vehicleModel
-String licensePlate
-String driverName
-String driverPhone
-boolean childSeat
-boolean wheelchairAccessible
-doulbe estimatedDistance
+calculateDuration() : long
+trackShuttle() : Location
+contacDriver() : void
+rateDriver() : boolean
<enumaration> ShuttleCompanies
nbktravel
tedblack
CitHopper
Morkel
Supershuttles
Shuttleandtours
extends
extends
genarates
Traveler
-Long travelerId
-int adultCount
-int childCount
-int infantCount
-List<String> travelerNames
-List<LocalDate> travelerAges
-List<String> passprtNumbers
+addAdult() : void
+removeAdult() : void
+addChild() : void
+removeChild() : void
+addInfant() : void
+removeInfant() : void
+validateTravelers() : void
+getTotalTravelers() : int
Invoice
-String InvoiceNumber
-LocalDateTime issueDate
-LocalDateTime dueDate
-Lost<LineItem> items
-double tax
-doulbe total
-String paymentTerms
+calculateTotal() : void
+generatePDF() : File
+sendViaEmail() : void
contains
LineItem
-String description
-Int quality
-double unitPrice
-double total
+calcalateTotal() : double
billed to
BillingAddress
-String fullName
-String addressLine1
-String addressLine2
-String city
-String state
-String postalCode
-String country
-String phone
has
CreditCardDetails
-String cardNumber
-String cardHolderName
-String expiryDate
-String lastFourDigits
-String cardType
-boolean isDefault
+validateCard() : boolean
+maskCardNmuber() : String
contains
<enumeration> PaymentStatus
PENDING
PAID
FAILED
REFUNDED
PARTIALLY_REFUNDED
HotleBooking
-String hoteId
-String hotelName
-String laction
-String hotalAddress
-String hotelPhone
-int starReting
-LocalDateTime checkIn
-LocalDateTime checkOut
-RoomTypeAvialable roomType
-RoomTypeByOccupancy occupancy
-RoomTypeByLayout layout
-RoomTypeByBedSize bedSize
-List<String> roomNumbers
-boolean breakFasIncluded
-boolean wifiIncluded
-boolean parkingIncluded
-List<String> specialRequests
-String cancellationPolicy
+calcutateNights() : long
+checkRoomAvailability() : boolean
+requestEarlyCheckIn() : boolean
+requestLateCheckOut() : boolean
+addSpecialRequest() : void
Uses
Uses
Uses
Uses
FlightBooking
-String flightNumber
-String airLine
-FJourney journeyType
-String fromLocation
-String toLocation
-LocalDateTime departureTime
-LocalDateTime arrivalTime
-String departureTerminal
-String arrivalTerminal
-FBookingClass bookingClass
-boolean isDirectFlight
-int stopOvers
-FlightType aircraftType
-String seatNumbers
-List<String> mealPreferences
-List<String> specialAssistance
-double baggageAllowence
-boolean travelInsurance
+checkIn() : BoardingPass
+selectSeat() : booloean
+addMealPreference() : void
+calculateFlightDuration() : long
+validateLuggage() : boolean
Uses*
<enumeration>FJourney
RETURN
ONE_WAY
MULTI_CITY
Uses
<enumeration> FlightType
AIRLINK
LIFT
FLYSAFEIR
CEMAIR
SOUTH_AFRICA_AIRWAYS
KENYE_AIRWAYS
Uses
<enumeration> FBookingClass
ECONOMY
PREMIUM_ECONOMY
BUSINESS
FIRST_CLASS
BookingPass
-String bookingRefrence
-String flightNumber
-String seatNumber
-String gate
-LocalDateTime boardingTime
-String qrCode
+genarate() : void
+validate() : boolean
<enumeration>
RoomTypeByBedSize
Queen
King
Twin
HollywoodTwin
Double-Double
Studio
<enumeration>
RoomTypeByLayout
Deluxe
SuitOrExecitive
MinSuiteOrJuniorSuite
Presidential
<enumeration>
RoomTypeByOccupancy
Single
Double
Triple
Quad
<enumeration>
RoomTypeAvailable
AccessibleRoom
RoomOnly
Pet_friendyRoom
Villa_Caban_Cable_ot_Bungalow
ExecutiveRoom
SmokingRoom
PaymentDetails
-Long paymentId
-String transactionId
-PaymentMethod method
-PaymentStatus status
-double amount
-String currency
-LocalDateTime paymentDate
-LocalaDateTime lastUpdated
-String receiptUrl
+processPayment() : boolean
+refund() : boolean
+generateReceipt() : Receipt
+validatePayment() : boolean
has
Uses
<enumeration> PaymentMethod
CREDIT_CARD
DEBIT_CARD
EFT
BANK_TRANSFER
Uses